Name: _____

Class: _____

Date: _____



Mr. Rawson • GCSE Physics

Kinematics 7 Extension

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A. Which equation? $a = \frac{v-u}{t}$ *or* $v^2 - u^2 = 2as$ *— the empty line in your Givens box tells you.*

1) Calculate the speed a go-kart was moving at before it accelerated at $2.5\,\text{m/s}^2$ for $4.0\,\text{s}$, reaching $18\,\text{m/s}$. **[3]**

Givens

$s =$
$u =$ **?**
$v = 18\ \text{m/s}$
$a = 2.5\ \text{m/s}^2$
$t = 4.0\ \text{s}$

No s, so $a = \frac{v-u}{t}$

$2.5 = \frac{18-u}{4.0} \quad \Rightarrow \quad 18 - u = 10 \quad \Rightarrow \quad u = \mathbf{8.0\ m/s}$

2) Calculate the acceleration of a motorbike that brakes from $24\,\text{m/s}$ to a stop in $48\,\text{m}$. **[3]**

Givens

$s = 48\ \text{m}$
$u = 24\ \text{m/s}$
$v = 0$
$a =$ **?**
$t =$

No t, so $v^2 - u^2 = 2as$

$v^2 - u^2 = 2 \quad a \quad s$

$0^2 - 24^2 = 2 \times a \times 48$

$$a = \frac{0^2 - 24^2}{2 \times 48} = \frac{-576}{96} = \mathbf{-6.0\ m/s^2}$$

3) Calculate the speed of a hockey puck after it slides $40\,\text{m}$ across the ice. It is hit at $5.0\,\text{m/s}$ and slows down with an acceleration of $-0.20\,\text{m/s}^2$. **[3]**

Givens

$s = 40\ \text{m}$
$u = 5.0\ \text{m/s}$
$v =$ **?**
$a = -0.20\ \text{m/s}^2$
$t =$

No t, so $v^2 - u^2 = 2as$

$v^2 - u^2 = 2 \quad a \quad s$

$v^2 - 5.0^2 = 2 \times (-0.20) \times 40$

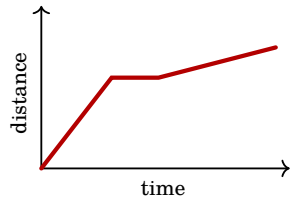
$v^2 = 5.0^2 + 2 \times (-0.20) \times 40 = 25 - 16 = 9.0$

$v = \sqrt{9.0} = \mathbf{3.0\ m/s}$

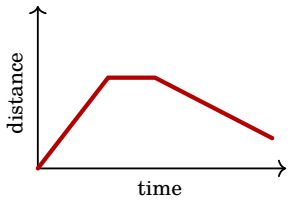
B. Graphs *choose the graph that matches.*

4) A bus drives at a steady speed, stops at a bus stop, then drives on **faster**. Which distance–time graph shows this journey? **[2]**

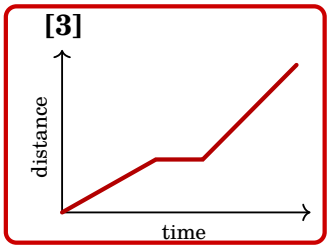
[1]



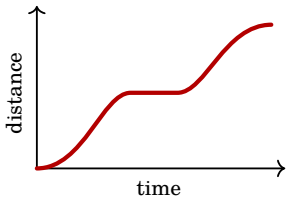
[2]



[3]



[4]

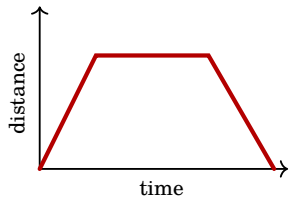


Graph **[3]** because

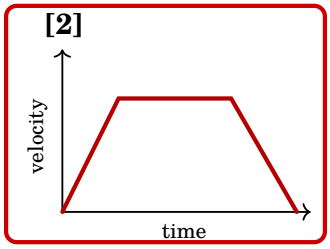
the line is flat (stationary) in the middle, and steeper after the stop than before it (faster).

5) A lift speeds up steadily from rest, moves at a constant speed, then slows steadily to a stop. Which graph shows this? **[1]**

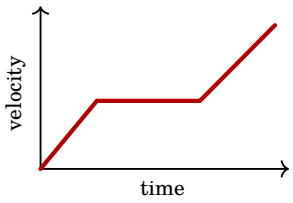
[1]



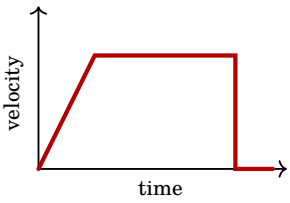
[2]



[3]

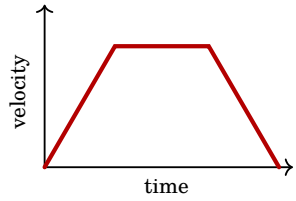


[4]

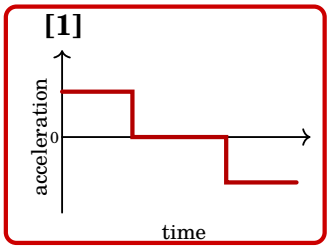


Graph **[2]** — [1] is a distance axis; [4] stops in no time at all.

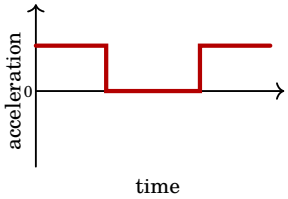
6) Here is a velocity–time graph. Which acceleration–time graph matches it? **[1]**



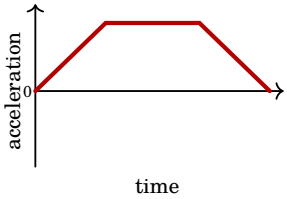
[1]



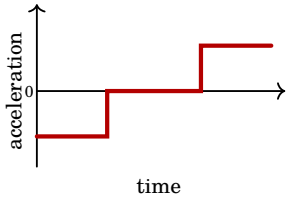
[2]



[3]

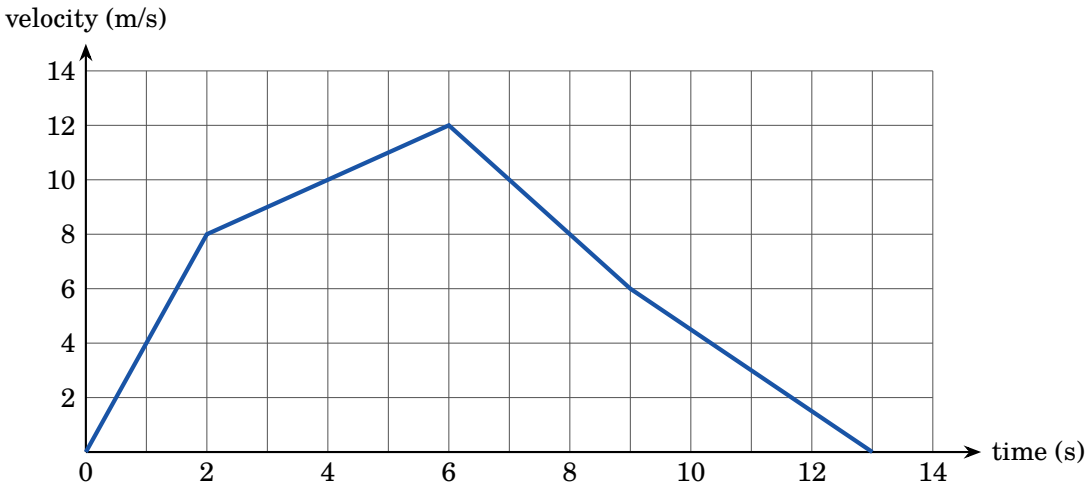


[4]



Graph **[1]** — gradient positive, then zero, then negative.

7) A motor boat leaves a jetty, then slows down to moor. The graph shows its motion.



(a) Between which times is the acceleration greatest? Explain how the graph shows this.

0 to 2 s — the line is steepest there (greatest gradient: 4.0 m/s^2).

[2]

(b) Calculate the acceleration between 2 s and 6 s.

$a = \dfrac{12-8}{6-2} = \dfrac{4}{4} = \mathbf{1.0\ m/s^2}$

[1]

(c) Describe the motion of the boat from 6 s to 13 s.

It slows down from 12 m/s to 6 m/s (deceleration 2.0 m/s^2), then slows more gently (1.5 m/s^2) until it stops at 13 s.

[2]

(d) Use $v^2 - u^2 = 2\,a\,s$ and your answer to (b) to calculate the distance travelled between 2 s and 6 s.

Givens

- $s =$ **?**
- $u = 8$ m/s
- $v = 12$ m/s
- $a = 1.0$ m/s^2
- $t = 4$ s

$v^2 - u^2 = 2 \quad a \quad s$

$12^2 - 8^2 = 2 \times 1.0 \times s$

$$s = \frac{12^2 - 8^2}{2 \times 1.0} = \frac{80}{2.0} = \mathbf{40\ m}$$

[2]

(e) Use the area under the graph to calculate the total distance the boat travels. The area between 2 s and 6 s matches your answer to (d).

$\frac{1}{2} \times 2 \times 8 \ + \ \frac{1}{2} \times (8+12) \times 4 \ + \ \frac{1}{2} \times (12+6) \times 3 \ + \ \frac{1}{2} \times 4 \times 6$

$= 8 + 40 + 27 + 12 =$ **87 m** *— the 2–6 s trapezium is 40 m, the same as (d).*

[3]